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AIME Combinatorics Problem Collection

Original mathematical restatements, 1983–2026. Contest credit: MAA American Mathematics Competitions. Refer to the linked source for official wording and diagrams. Answers and worked solutions are in the separate solution collection.

1983 AIME Problem 7: Adjacent knights

Three of 25 labeled seats around a circle are selected uniformly. For the probability that a selected pair is adjacent, find the sum of numerator and denominator in lowest terms.

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1983 AIME Problem 10: One repeated decimal digit

Count four-digit integers beginning with 1 whose digit multiplicities are exactly 2,1,1.

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1983 AIME Problem 13: Sum of alternating subset sums

For every nonempty subset of 1 through 7, list its elements decreasingly and alternately add and subtract them, starting with plus. Sum these values over all subsets.

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1984 AIME Problem 10: A score that determines the number correct

A 30-question test scores 30+4c−w, with c correct, w wrong and the rest blank. Find the smallest score above 80 that uniquely determines c.

Published problem and solutions · MAA AMC.

1984 AIME Problem 11: Separated birches

A row contains three maples, four oaks and five birches, arranged uniformly. Find numerator plus denominator for the probability that no birches are adjacent.

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1984 AIME Problem 12: Root orbits under two reflections

A real function is unchanged by reflection about 2 and about 7, and vanishes at 0. Find the fewest roots it can have in [−1000,1000].

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1985 AIME Problem 8: Balanced integer rounding

Round 2.56,2.61,2.65,2.71,2.79,2.82,2.86 to integers with total 19, minimizing the largest absolute error M. Find 100M.

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1985 AIME Problem 10: Values of a floor sum

How many integers from 1 through 1000 occur as floor(2x)+floor(4x)+floor(6x)+floor(8x) for real x?

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1985 AIME Problem 12: Random walk on a tetrahedron

A bug starts at one vertex of a unit-edge tetrahedron and repeatedly chooses each incident edge with probability 1/3. Its chance of returning after seven moves is n/729. Find n.

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1985 AIME Problem 14: Tournament score balance

Every player in a round robin earns half their points against the ten lowest-scoring players; wins score 1 and draws 1/2 each. Determine the total number of players.

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1986 AIME Problem 7: Ranking subsets by ternary value

Arrange all positive sums of distinct nonnegative powers of 3 in increasing order. Find the hundredth term.

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1986 AIME Problem 10: Recovering a permuted numeral

The five digit permutations of a three-digit numeral other than its original ordering have sum 3194. Recover the original numeral, allowing leading zeros in rearrangements.

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1986 AIME Problem 12: A maximum set with distinct subset sums

Choose a subset S of {1,...,15} so that no two disjoint nonempty subsets have equal sums. Maximize the sum of S.

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1986 AIME Problem 13: Prescribed adjacent transitions

Count 15-letter H/T strings with exactly 2 HH, 3 HT, 4 TH and 5 TT adjacent pairs.

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1987 AIME Problem 1: Addition without carrying

Count ordered nonnegative integer pairs (m,n) summing to 1492 with no decimal carry.

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1987 AIME Problem 7: LCM triples

Count positive ordered triples (a,b,c) with lcm(a,b)=1000 and lcm(a,c)=lcm(b,c)=2000.

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1987 AIME Problem 13: One bubble pass

A random ordering of 40 distinct numbers undergoes one left-to-right bubble-sort pass. Find numerator plus denominator for the chance that the original twentieth number finishes thirtieth.

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1988 AIME Problem 1: More lock combinations

A ten-button lock originally accepts any five-element button set as a possible code. If codes of every size 1 through 9 become available, how many additional codes exist?

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1988 AIME Problem 5: Divisor probability

A divisor of 10^99 is chosen uniformly. Find numerator plus denominator for its probability of being divisible by 10^88.

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1988 AIME Problem 10: Interior diagonals of a polyhedron

A convex polyhedron has 12 square, 8 regular hexagonal and 6 regular octagonal faces, one of each at every vertex. Count vertex-pair segments lying in its interior.

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1988 AIME Problem 15: Stack output after lunch

Letters 1 through 9 arrive in order at a last-in-first-out inbox. At lunch, letter 8 has already been typed. Count distinct possible sequences of letters typed afterward, including the empty sequence.

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1989 AIME Problem 2: Polygons from circle points

Ten points lie on a circle. Count convex polygons with at least three vertices selected from those points.

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1989 AIME Problem 5: Recovering a biased coin

For five independent flips of a fixed biased coin, the positive probabilities of exactly one and exactly two heads agree. Find numerator plus denominator for the probability of exactly three heads.

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1989 AIME Problem 11: Largest mode-minus-mean

A sample of 121 integers in [1,1000] has one unique most frequent value. Maximize the integer part of that value minus the sample mean.

Published problem and solutions · MAA AMC.

1989 AIME Problem 13: Avoiding differences four and seven

Find the maximum size of a subset of {1,...,1989} with no pair differing by 4 or 7.

Published problem and solutions · MAA AMC.

1990 AIME Problem 1: Ranking nonsquares and noncubes

List positive integers that are neither perfect squares nor perfect cubes. Find the five-hundredth.

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1990 AIME Problem 6: Tagged fish population estimate

Of 60 fish tagged in May, 25% are assumed gone by September. A representative September sample of 70 contains three tagged fish; 40% of September fish are new. Estimate the May population.

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1990 AIME Problem 8: Interleaving three target stacks

Targets occupy three labeled vertical columns of sizes 3,3,2. Each shot must hit the lowest remaining target of its chosen column. Count complete shooting orders.

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1990 AIME Problem 9: No consecutive heads

Toss a fair coin ten times. Find numerator plus denominator for the probability of no adjacent heads.

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1990 AIME Problem 12: Sum of dodecagon chords

For a regular 12-gon of circumradius 12, write the sum of all side and diagonal lengths as a+b√2+c√3+d√6. Find a+b+c+d.

Published problem and solutions · MAA AMC.

1990 AIME Problem 13: Leading nines in powers

Among $9^{0,\cdots,9}4000$, count those beginning with decimal digit 9, given that 9^4000 has 3817 digits and begins with 9.

Published problem and solutions · MAA AMC.

1991 AIME Problem 5: Coprime factor allocations

How many reduced fractions strictly between 0 and 1 have numerator times denominator equal to 20!?

Published problem and solutions · MAA AMC.

1991 AIME Problem 10: Lexicographic transmission errors

Send aaa and bbb through independent channels that flip each letter with probability 1/3. The first received word precedes the second with probability p. Find the numerator of p in lowest terms.

Published problem and solutions · MAA AMC.

1991 AIME Problem 13: Sock counts from a probability

At most 1991 socks, red or blue, are in a drawer. Two drawn without replacement have the same color with probability 1/2. Maximize the number of red socks.

Published problem and solutions · MAA AMC.

1992 AIME Problem 2: Strictly increasing digits

Count positive decimal integers of at least two digits whose digits strictly increase from left to right.

Published problem and solutions · MAA AMC.

1992 AIME Problem 5: Numerators of repeating decimals

For all numbers 0.abcabc… strictly between 0 and 1, count distinct numerators after reducing the fraction.

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1992 AIME Problem 6: Consecutive integers without carries

Count consecutive pairs whose members are both in [1000,2000] and whose decimal addition produces no carry.

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1992 AIME Problem 12: Chomp positions

On a 5-by-7 board, a move removes one cell and all cells above and to its right. Count all possible remaining cell sets, including the full and empty boards.

Published problem and solutions · MAA AMC.

1992 AIME Problem 15: Missing factorial trailing-zero counts

A positive integer is attainable if it is the exact number of trailing zeros of some factorial. How many positive integers below 1992 are unattainable?

Published problem and solutions · MAA AMC.

1993 AIME Problem 1: Even numbers with distinct digits

Count even integers strictly between 4000 and 7000 whose four digits are distinct.

Published problem and solutions · MAA AMC.

1993 AIME Problem 4: Equal outer and inner sums

Count integer quadruples $0<a<b<c<d<500$ satisfying $a+d=b+c$ and $bc-ad=93$.

Published problem and solutions · MAA AMC.

1993 AIME Problem 7: Fitting one random brick into another

Choose six distinct values uniformly from 1 through 1000, three for brick dimensions and three for box dimensions. Find numerator plus denominator for the probability that the brick fits with parallel sides after rotation.

Published problem and solutions · MAA AMC.

1993 AIME Problem 8: Unordered covering subset pairs

For a six-element set S, count unordered pairs of subsets, allowing equality, whose union is S.

Published problem and solutions · MAA AMC.

1993 AIME Problem 10: Uniform vertex face incidences

A convex polyhedron has 32 triangular or pentagonal faces. At each vertex, T triangles and P pentagons meet. Determine $100P+10T+V$.

Published problem and solutions · MAA AMC.

1993 AIME Problem 11: Loser starts the next game

Two players alternate fair coin flips until the first head wins. The loser starts the next game. Alfred starts game one. If his probability of winning game six is m/n in lowest terms, find the last three digits of m+n.

Published problem and solutions · MAA AMC.

1993 AIME Problem 12: Undoing random midpoint choices

Inside the triangle with vertices (0,0),(0,420),(560,0), a point undergoes six midpoint moves toward arbitrarily selected vertices and ends at (14,92). Determine the sum of its starting coordinates.

Published problem and solutions · MAA AMC.

1994 AIME Problem 5: A weighted sum over digit strings

For each positive integer n, multiply its nonzero decimal digits. Let S be the sum of these products for n=1,…,999. Find the largest prime divisor of S.

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1994 AIME Problem 6: Small triangles in three line families

For every integer k from -10 to 10, draw y=k, $y=\sqrt{3}x+2k$ and $y=-\sqrt{3}x+2k$. Count equilateral triangles of side $2/\sqrt{3}$ whose edges lie on the lines.

Published problem and solutions · MAA AMC.

1994 AIME Problem 7: Lines with only integral circle intersections

Count real pairs (a,b) for which $ax+by=1$ and $x^2+y^2=50$ have at least one common point and every common point has integral coordinates.

Published problem and solutions · MAA AMC.

1994 AIME Problem 9: Matched tiles without three open pairs

Six labeled pairs of tiles are drawn uniformly without replacement. A pair is discarded as soon as matched. The process loses if three unmatched tiles are ever held. Find numerator plus denominator for the probability of emptying the bag.

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1994 AIME Problem 11: Distinct heights of a brick tower

Build a tower of 94 bricks, each contributing height 4,10 or 19. Count distinct possible total heights.

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1995 AIME Problem 3: First hitting a lattice point

A planar random walk starts at (0,0) and chooses each unit cardinal step with probability 1/4. Find numerator plus denominator for the chance it visits (2,2) within six steps.

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1995 AIME Problem 6: Divisors below a square root

Let n=$2^{31} \cdot 3$19. Count divisors of n² that are smaller than n but do not divide n.

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1995 AIME Problem 8: Two simultaneous integer ratios

Count positive integer pairs $y < x \leq 100$ for which x/y and $(x+1)/(y+1)$ are integers.

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1995 AIME Problem 11: A self-similar cut prism

An integer-sided rectangular prism has sorted sides $a \leq b \leq c$ with $b = 1995$. A face-parallel cut produces a smaller similar prism. Count possible triples (a, b, c).

Published problem and solutions · MAA AMC.

1995 AIME Problem 15: Five heads before two tails

Flip a fair coin until either five consecutive heads or two consecutive tails first appears. Find numerator plus denominator for the probability that the five-head run wins.

Published problem and solutions · MAA AMC.

1996 AIME Problem 2: Even logarithm floors

Count positive integers $n < 1000$ for which $\lfloor \log_2 n \rfloor$ is positive and even.

Published problem and solutions · MAA AMC.

1996 AIME Problem 3: Counting monomials after expansion

Find the least positive n for which (xy−3x+7y−21)^n has at least 1996 nonzero monomials after collecting terms.

Published problem and solutions · MAA AMC.

1996 AIME Problem 6: No perfect or winless team

Five teams play every pair once, with equally likely independent outcomes. Find numerator plus denominator for the probability that each team both wins and loses at least once.

Published problem and solutions · MAA AMC.

1996 AIME Problem 7: Two marked squares up to rotation

Two cells of a 7-by-7 board are yellow and the rest green. Count colorings up to rotations of the board, with reflections remaining distinct.

Published problem and solutions · MAA AMC.

1996 AIME Problem 8: Integer pairs of prescribed harmonic mean

Count positive integer pairs x<y whose harmonic mean is 6^20.

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1996 AIME Problem 9: Alternating elimination along lockers

Initially lockers 1,…,1024 are closed. On alternating left-to-right and right-to-left passes, open every other still-closed locker, starting with the first encountered. Find the last opened locker.

Published problem and solutions · MAA AMC.

1996 AIME Problem 12: Mean sum of five pair distances

Permute 1,…,10 uniformly, pair adjacent entries in five disjoint pairs, and add their absolute differences. If the mean is p/q in lowest terms, find p+q.

Published problem and solutions · MAA AMC.

1996 AIME Problem 14: Cubes crossed by a space diagonal

A box of side lengths 150,324,375 is partitioned into unit cubes. Count cubes whose interiors meet a long diagonal.

Published problem and solutions · MAA AMC.

1997 AIME Problem 1: Differences of squares

Count integers 1 through 1000 that are differences of squares of nonnegative integers.

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1997 AIME Problem 2: Squares among grid rectangles

Let r be the number of rectangles and s the number of squares formed by the nine horizontal and nine vertical lines of a standard 8-by-8 board. Reduce s/r=m/n and find m+n.

Published problem and solutions · MAA AMC.

1997 AIME Problem 5: Decimal numbers closest to two sevenths

Count four-place decimals 0.abcd for which 2/7 is the closest fraction whose numerator is 1 or 2 and whose denominator is an integer.

Published problem and solutions · MAA AMC.

1997 AIME Problem 8: Balanced sign arrays

Count 4-by-4 arrays of ±1 in which every row and column sums to zero.

Published problem and solutions · MAA AMC.

1997 AIME Problem 10: Complementary attribute triples

A deck has all 27 combinations of three attributes, each with three values. Count three-card sets in which each attribute is either constant across the cards or takes all three values.

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1997 AIME Problem 14: A short arc probability

Two distinct 1997th roots of unity are chosen uniformly. Find numerator plus denominator for the chance that $|v+w| \geq \sqrt{2+\sqrt{3}}$.

Published problem and solutions · MAA AMC.

1998 AIME Problem 1: Counting possible LCM partners

How many positive integers k satisfy lcm(6^6,8,k)=12^12?

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1998 AIME Problem 2: Integer points between two slope bounds

Count positive integer pairs with $x \leq 2y \leq 60$ and $y \leq 2x \leq 60$.

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1998 AIME Problem 4: Three odd hand sums

Randomly divide tiles numbered 1,…,9 among three labeled players, three tiles each. Find numerator plus denominator for the probability that all three hand sums are odd.

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1998 AIME Problem 7: Odd positive compositions

Let n count ordered quadruples of positive odd integers summing to 98. Find n/100.

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1998 AIME Problem 9: Overlapping coffee breaks

Two people arrive independently and uniformly during one hour and each stays m minutes. Their visits overlap with probability 40%. Write m=a−b√c with c squarefree and find a+b+c.

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1998 AIME Problem 13: Complex weights over subsets

For each nonempty subset of {1,…,n}, list its elements increasingly and weight the jth by i^j. Let S_n sum these weighted subset sums. Given S_8=−176−64i and S_9=p+qi, find |p|+|q|.

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1998 AIME Problem 15: Longest domino chain

A chain uses distinct ordered pairs (i, j), $1 \le i, j \le 40$, $i \ne j$, with successive pairs sharing the connecting endpoint. It may not use both (i, j) and (j, i). Maximize its length.

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1999 AIME Problem 5: Possible digit-sum changes

Let S(x) be the decimal digit sum and T(x)=|S(x+2)−S(x)| for positive integers x. Count distinct T-values at most 1999.

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1999 AIME Problem 7: Four-position divisor switches

Switches labeled $2^x 3^y 5^z$ for $0 \le x, y, z \le 9$ cycle $A \to B \to C \to D \to A$. Processing every label once advances all switches whose labels divide it. Starting at A, how many finish at A?

Published problem and solutions · MAA AMC.

1999 AIME Problem 8: Exactly two thresholds on a simplex

On the nonnegative triangle $x + y + z = 1$, take points satisfying exactly two of $x \ge 1/2, y \ge 1/3, z \ge 1/6$. If their area fraction is m/n reduced, find $m + n$.

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1999 AIME Problem 10: A triangle among four chosen segments

From the 45 segments joining ten points with no three collinear, choose four uniformly. Find numerator plus denominator for the probability that three form a triangle on the given points.

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1999 AIME Problem 13: Distinct tournament win totals

Forty teams play each pair once with equally likely independent winners. If the probability that all win totals differ is m/n in lowest terms, find $\log_2 n$.

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2000 AIME I Problem 3: Adjacent binomial coefficients

For coprime positive integers a, b, the coefficients of x^2 and x^3 in $(ax + b)^{2000}$ agree. Determine $a + b$.

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2000 AIME I Problem 5: Two urns with a fixed total

Two urns each contain both black and white balls, with 25 balls in total. Independent uniform draws, one per urn, are both black with probability 27/50. Write the probability of two whites as reduced m/n and find $m + n$.

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2000 AIME I Problem 11: Independent prime exponents

Sum a/b over ordered pairs of coprime positive divisors of 1000. If the sum is S, find $\lfloor S/10 \rfloor$.

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2000 AIME I Problem 12: Reflection orbits

A real function satisfies $f(x) = f(398 - x) = f(2158 - x) = f(3214 - x)$ for every real x. What is the largest number of different values among $f(0), \ldots, f(999)$?

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2000 AIME I Problem 15: Alternating elimination

A stack of 2000 labeled cards is processed by removing the top card, then moving the next card to the bottom, repeatedly. Removed labels are $1, 2, \ldots, 2000$. How many cards originally preceded label 1999?

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2000 AIME II Problem 3: A second matching pair

A deck has four cards of each rank $1, \ldots, 10$. Remove two cards of one rank. Two further cards are sampled uniformly without replacement. If their matching probability is m/n in lowest terms, find $m + n$.

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2000 AIME II Problem 5: Ordered rings on fingers

Choose five of eight distinct rings and distribute them on four labeled fingers. Their vertical order matters; empty fingers are allowed. Find the first three nonzero digits of the number of arrangements.

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2000 AIME II Problem 7: Half a binomial row

Define N by $\sum_{k=2}^{9} 1/(k!(19-k)!) = N/18!$. Find the greatest integer strictly smaller than $N/100$.

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2001 AIME I Problem 6: Sorted dice

Four independent fair six-sided dice rolls are nondecreasing in time with probability m/n in lowest terms. Find $m + n$.

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2001 AIME I Problem 10: Parity classes of lattice points

Choose two distinct points uniformly from $\{0, 1, 2\} \times \{0, 1, 2, 3\} \times \{0, 1, 2, 3, 4\}$. Their midpoint belongs to the same set with probability m/n in lowest terms. Find $m + n$.

Published problem and solutions · MAA AMC.

2001 AIME I Problem 11: Two numberings of a rectangular array

In a $5 \times N$ array choose one point per row. Row-major labels x_i and column-major labels y_i obey $x_1 = y_2, x_2 = y_1, x_3 = y_4, x_4 = y_5, x_5 = y_3$. Find the smallest N.

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2001 AIME I Problem 14: Binary strings with two forbidden runs

Count binary strings of length 19 containing neither 11 nor 000.

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2001 AIME I Problem 15: Hamilton cycles in a cube complement

Label the eight faces of a regular octahedron randomly by $1, \ldots, 8$. No consecutive labels, including 8, 1, may share an edge. Express the probability as reduced m/n and find $m + n$.

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2001 AIME II Problem 1: Digit adjacency as a directed graph

Find the leading three digits of the largest positive integer whose every adjacent two-digit block is a perfect square.

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2001 AIME II Problem 2: Bounds on an intersection

Among 2001 students everyone studies Spanish or French. Between 80% and 85% study Spanish and between 30% and 40% study French. Find the difference between the maximum and minimum possible numbers studying both.

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2001 AIME II Problem 5: Triangle-free ordered sets

Every ten-element subset of $\{4, 5, \ldots, n\}$ must contain three side lengths of a nondegenerate triangle. Find the largest such n.

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2001 AIME II Problem 9: Avoiding red subsquares

Color the nine cells of a 3×3 board independently red or blue, equally likely. Express the probability of no all-red 2×2 subsquare as reduced m/n and find $m + n$.

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2001 AIME II Problem 10: Counting exponent gaps

How many different numbers $10^j - 10^i$, with $0 \le i < j \le 99$, are divisible by 1001?

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2001 AIME II Problem 11: Wins versus losses

A team independently wins, draws, or loses each of six games, with all three outcomes equally likely. Write the probability of more wins than losses as reduced m/n and find $m+n$.

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2001 AIME II Problem 12: Recursive tetrahedral growth

Begin with a unit-volume regular tetrahedron. At each stage replace the middle quarter-triangle of every triangular face by the three exposed faces of an outward regular tetrahedron. Find numerator plus denominator of the volume after three stages.

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2002 AIME I Problem 1: A palindrome in either block

A uniformly random plate consists of three letters from a 26-letter alphabet followed by three decimal digits. Either block may be a palindrome. Find numerator plus denominator of the probability that at least one is.

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2002 AIME I Problem 3: Reversed two-digit ages

One person is currently 25, and another is older. In a positive integer number n of years their ages are two-digit reversals. Count the possible ordered pairs consisting of the older person's present age and n.

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2002 AIME I Problem 5: Squares through dodecagon vertices

How many squares in the plane have at least two vertices among the vertices of a fixed regular dodecagon?

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2002 AIME I Problem 9: Three arithmetic progressions partition the integers

Three progressions start at $1, 2, 3$ and have positive integer steps h, t, u. They partition the positive integers exactly once. Sum $100h + 10t + u$ over all valid triples.

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2002 AIME I Problem 14: Integer means after deletion

A set of distinct positive integers contains 1, has maximum 2002, and has integral mean after deleting any single member. Find its largest possible size.

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2002 AIME II Problem 9: Unordered disjoint nonempty subsets

Count unordered pairs of nonempty disjoint subsets of a ten-element set; give the remainder modulo 1000.

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2002 AIME II Problem 12: A rational ballot boundary

Independent shots succeed with probability $2/5$. Require exactly four successes in ten shots and a success fraction at most $2/5$ after every earlier shot. Express the probability as $p^a q^b r/s^c$, with the stated factors prime, and find $(p+q+r+s)(a+b+c)$.

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2003 AIME I Problem 3: Charge each pair to its maximum

For each two-element subset of $\{1, 2, 3, 5, 8, 13, 21, 34\}$, record its larger member. Find the sum of the recorded numbers.

Published problem and solutions · MAA AMC.

2003 AIME I Problem 6: Triangle types in a cube

Sum the areas of all triangles with vertices among the eight vertices of a unit cube. Write the sum as $m + \sqrt{n} + \sqrt{p}$ with integer parameters and find $m + n + p$.

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2003 AIME I Problem 9: Equal digit sums

Count four-digit decimal integers whose first two digits have the same sum as their last two digits.

Published problem and solutions · MAA AMC.

2003 AIME I Problem 11: Triangle inequalities as interval probability

Choose x uniformly from $(0, 90^\circ)$. The probability that $\sin^2 x, \cos^2 x, \sin x \cos x$ fail to form a triangle is d/n, where d is the degree measure of $\arctan m$ and $m + n < 1000$. Find $m + n$.

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2003 AIME I Problem 13: A majority of binary ones

Let N count integers from 1 through 2003 whose binary expansions contain more ones than zeros. Find $N \bmod 1000$.

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2003 AIME II Problem 3: Uniform outdegree words

Count length-seven words on $\{A, B, C\}$ that avoid the adjacent blocks AB, BC, CA.

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2003 AIME II Problem 12: Minimum occupancy with percentage constraints

Twenty-seven candidates receive integer vote counts totaling s. For each candidate, its vote count exceeds its exact percentage of all votes by at least one. Find the smallest s.

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2003 AIME II Problem 13: Return walk on a triangle

A walker starts at a vertex of a triangle and at each step chooses uniformly one of the other two vertices. Find numerator plus denominator of its probability of returning after ten steps.

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2004 AIME I Problem 3: Pairs of polyhedron vertices

A convex polyhedron has 26 vertices, 60 edges, 24 triangular faces and 12 quadrilateral faces. Count pairs of vertices whose segment is neither an edge nor contained in a face.

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2004 AIME I Problem 6: Alternating four-digit permutations

Count four-digit integers with distinct digits satisfying $a_1 < a_2 > a_3 < a_4$.

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2004 AIME I Problem 7: Pair contributions to a coefficient

Find the absolute value of the coefficient of x^2 in $\prod_{k=1}^{15}(1+(-1)^k kx)$.

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2004 AIME I Problem 8: Regular star polygons

Count similarity classes of connected regular 1000-vertex star polygons with equal edge lengths and equal counterclockwise turning angles less than 180°, requiring a genuine edge crossing.

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2004 AIME I Problem 10: A random disk avoids a diagonal

A unit-radius disk has its center uniformly distributed among positions where it fits in a 15×36 rectangle. Find numerator plus denominator of the probability that it misses a fixed diagonal.

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2004 AIME I Problem 15: Reverse dynamics and forbidden operation runs

Iterate $f(1)=1$, $f(x)=x/10$ when $10 \mid x$, and $f(x)=x+1$ otherwise. Let $d(x)$ be the first index with value 1, starting at index one. If m integers have $d(x)=20$, find the sum of distinct prime factors of m.

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2004 AIME II Problem 2: Matching unordered color profiles

From ten red and ten blue candies, one person draws two and then a second draws two without replacement. The probability that their unordered color combinations agree is reduced m/n. Find $m+n$.

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2004 AIME II Problem 3: Hidden cubes

A rectangular block made of unit cubes has exactly 231 cubes invisible when three mutually adjacent faces are viewed. Find the least possible total number of cubes.

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2004 AIME II Problem 4: At most two decimal symbols

Count positive integers below 10000 whose decimal expansions use at most two distinct digits.

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2004 AIME II Problem 8: Divisor counts as exponent allocations

How many divisors of 2004^{2004} have exactly 2004 positive divisors?

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2004 AIME II Problem 10: Two binary ones modulo nine

Choose uniformly an integer below 2^{40} with exactly two binary ones. Find numerator plus denominator of the probability that it is divisible by 9.

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2004 AIME II Problem 14: Distinct digit lengths of a sum

A string of n sevens is split into decimal integers by inserting plus signs. How many values of n can give a total of 7000?

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2004 AIME II Problem 15: Track a layer through folds

A strip of 1024 numbered unit squares is repeatedly folded right half over left until one stack remains. How many squares lie below the square originally numbered 942 from the left?

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2005 AIME I Problem 5: Coin colors and orientations separate

Stack four identical gold coins and four identical silver coins. Each has a marked face, and adjacent marked faces may not face each other. Count distinct stacks, including orientations.

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2005 AIME I Problem 9: Local exposure types in a random cube

Each of 27 unit cubes has four painted faces, with its two unpainted faces adjacent. Orient them independently uniformly in a $3 \times 3 \times 3$ cube. If the probability that the outer surface is painted is $p^a/(q^b r^c)$ in prime form, find $a+b+c+p+q+r$.

Published problem and solutions · MAA AMC.

2005 AIME I Problem 12: Parity changes only at squares

Let $S(n) = \sum_{k=1}^{n} \tau(k)$, where τ counts positive divisors. Among $1 \le n \le 2005$, let a and b count odd and even values of $S(n)$. Find $|a - b|$.

Published problem and solutions · MAA AMC.

2005 AIME I Problem 13: Diagonal separators in lattice paths

Count paths from $(0,0)$ to $(5,5)$ with steps $(1,0), (0,1), (1,1)$ and with no consecutive horizontal and vertical steps in either order.

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2005 AIME II Problem 1: A binomial ratio determines deck size

A deck with $n \ge 6$ distinct cards has six times as many six-card subsets as three-card subsets. Find n.

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2005 AIME II Problem 2: Three balanced meal bags

There are three rolls of each of three types. Randomly divide them into three labeled bags of size three. Find numerator plus denominator of the probability that every bag contains all three types.

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2005 AIME II Problem 4: Union of divisor sets

Count positive integers dividing at least one of $10^{10}, 15^7, 18^{11}$.

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2005 AIME II Problem 6: Fixed points of a shuffle

A stack labeled $1, \ldots, 2n$ top to bottom is cut into two equal halves. Restack by alternately placing the next card from the lower half and then the upper half on top. At least one card from each half stays fixed, including card 131. Find $2n$.

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2006 AIME I Problem 2: No gaps among fixed-size subset sums

How many different sums can a 90-element subset of $\{1, \ldots, 100\}$ have?

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2006 AIME I Problem 6: Uniform digit contributions

Sum all repeating decimals $0.\overline{abc}$ with three distinct decimal digits, allowing $a = 0$.

Published problem and solutions · MAA AMC.

2006 AIME I Problem 11: Insert the largest block

Permute blocks of sizes $1, \ldots, 8$ bottom to top, requiring every size to be at most two greater than the size immediately below. Find the count modulo 1000.

Published problem and solutions · MAA AMC.

2006 AIME I Problem 13: Valuation layers

For even x, let $g(x)$ be the largest power of two dividing x. Set $S_n = \sum_{k=1}^{2^{n-1}} g(2k)$. Find the greatest $n < 1000$ for which S_n is a square.

Published problem and solutions · MAA AMC.

2006 AIME I Problem 15: A telescoping potential

Start with $x_0 = 0$ and choose x_k subject to $|x_k| = |x_{k-1} + 3|$. Find the minimum of $|x_1 + \cdots + x_{2006}|$.

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2006 AIME II Problem 4: A permutation with one valley

Count permutations of $1, \ldots, 12$ satisfying $a_1 > \cdots > a_6 < a_7 < \cdots < a_{12}$.

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2006 AIME II Problem 5: Opposite-face bias

A die assigns probability $1/6 + t$ to a face, $1/6 - t$ to its opposite, and $1/6$ to the four others; opposite labels sum to seven. Two independent such dice sum to seven with probability $47/288$. Find numerator plus denominator of the larger face probability.

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2006 AIME II Problem 7: Complementary decimal numbers

Count ordered positive integer pairs (a, b) with $a + b = 1000$ and no zero digit in either number.

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2006 AIME II Problem 8: Three corners modulo all permutations

Build an equilateral triangle from four congruent small equilateral triangles. Each may have one of six colors. Identify configurations under rotations and reflections. Count the resulting colorings.

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2006 AIME II Problem 10: A one-point head start

Seven teams play each other once, with independent fair win-loss outcomes. Given that A defeats B, find numerator plus denominator of the probability that A finishes with more wins than B.

Published problem and solutions · MAA AMC.

2006 AIME II Problem 14: Reciprocal digit frequency

Let S_n be the sum of reciprocals of all nonzero decimal digits appearing in $1, \ldots, 10^n$. Find the least positive n for which S_n is integral.

Published problem and solutions · MAA AMC.

2007 AIME I Problem 5: A periodic rounding map

Convert an integer Fahrenheit temperature in $[32, 1000]$ to Celsius using $C = 5(F-32)/9$, round to the nearest integer, convert back, and round again. Count temperatures recovered exactly.

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2007 AIME I Problem 6: Next-multiple walks

From an integer position a frog jumps to the next larger multiple of 3 or of 13. Count distinct coordinate sequences from 0 to 39.

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2007 AIME I Problem 10: Binary matrices with fixed margins

A 6×4 binary matrix has two ones in every row and three in every column. Find the number of such matrices modulo 1000.

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2007 AIME I Problem 11: Rounding square roots in blocks

For each positive integer p, let $b(p)$ be its square root rounded to the nearest integer. Find $\sum_{p=1}^{2007} b(p) \pmod{1000}$.

Published problem and solutions · MAA AMC.

2007 AIME II Problem 1: A repeated symbol in a license string

Form five-character strings from the multiset $\{A, I, M, E, 2, 0, 0, 7\}$ without exceeding any multiplicity. If the count is N, find $N/10$.

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2007 AIME II Problem 2: Divisibility becomes a composition

Count ordered positive triples (a, b, c) with $a \mid b$, $a \mid c$, and $a + b + c = 100$.

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2007 AIME II Problem 5: Grid cells below a rational line

Count unit grid squares whose interiors lie in the first quadrant and below $9x + 223y = 2007$.

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2007 AIME II Problem 6: Uniform indegrees in a digit graph

Count four-digit integers such that an odd digit is followed by a larger digit and an even digit by a smaller digit.

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2007 AIME II Problem 8: Optimizing a rectangular grid

Draw full lines parallel to the sides of a 4×5 sheet, with total drawn length 2007. Basic rectangles must have four drawn sides and no drawn line inside. Find the maximal number modulo 1000.

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2007 AIME II Problem 10: Containment in a set or its complement

Independently choose uniform subsets A, B of a six-element set. The probability that $B \subseteq A$ or $B \subseteq A^c$ is m/n^r in reduced prime-denominator form. Find $m + n + r$.

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2007 AIME II Problem 13: Pascal propagation modulo three

Place eleven binary digits in the bottom row of a triangular array. Each upper entry is the sum of its two children. Count assignments making the top entry divisible by 3.

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2008 AIME I Problem 6: Summing an arithmetic row repeatedly

The top row is $1, 3, \ldots, 99$. Each row below sums adjacent entries until one remains. Count all entries divisible by 67.

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2008 AIME I Problem 7: Occupied hundred-blocks

For $0 \le i \le 999$, let $S_i = \{100i, \ldots, 100i + 99\}$. How many of these sets contain no perfect square?

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2008 AIME I Problem 9: Random crate heights

Ten independent crates have heights $3, 4, 6$ with equal probabilities. The probability their total height is 41 is reduced m/n. Find m.

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2008 AIME I Problem 11: Parity restrictions on runs

Count length-14 words on $\{A, B\}$ with every maximal A-run even and every maximal B-run odd.

Published problem and solutions · MAA AMC.

2008 AIME II Problem 10: Increasing distances on a small grid

On the sixteen points $\{0,1,2,3\}^2$, a path visits distinct points with strictly increasing successive distances. Let m be its maximum number of points and r the number of paths attaining that maximum. Find mr.

Published problem and solutions · MAA AMC.

2008 AIME II Problem 12: A separator that behaves like a blue flag

Distribute ten identical blue and nine identical green flags onto two labeled nonempty flagpoles, with vertical order relevant and no adjacent greens on either pole. Find the arrangement count modulo 1000.

Published problem and solutions · MAA AMC.

2009 AIME I Problem 3: Cancel symmetric binomial coefficients

Eight independent coin flips have head probability p. The chance of three heads is one twenty-fifth of the chance of five heads. Find numerator plus denominator of p in lowest terms.

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2009 AIME I Problem 8: Sum of pairwise differences

Sum all positive differences between pairs from $\{1,2,4,\ldots,2^{10}\}$, and report the remainder modulo 1000.

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2009 AIME I Problem 9: Three prices from seven digits

Form an ordered triple of positive prices at most 9999 using exactly four digits 1 and three digits 3 in total. Count possible triples.

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2009 AIME I Problem 10: Cyclic species runs

Arrange five distinct members of each of three species around fifteen numbered seats. Seat 1 is M, seat 15 is E, and allowed changes clockwise are only $M \to V \to E \to M$. If the count is $N(5!)^3$, find N.

Published problem and solutions · MAA AMC.

2009 AIME I Problem 11: Integral triangle areas on a line

Choose distinct nonnegative lattice points P, Q on $41x + y = 2009$. Count triangles OPQ of positive integer area, where $O = (0,0)$.

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2009 AIME II Problem 6: At least one consecutive pair

Count five-element subsets of $\{1,\ldots,14\}$ containing consecutive integers, and give the residue modulo 1000.

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2009 AIME II Problem 8: Nearby geometric stopping times

Two people independently roll fair dice until the first six. Find numerator plus denominator of the probability that their numbers of rolls differ by at most one.

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2009 AIME II Problem 9: An injection leaves two boundary slices

Let m, n count positive integer solutions to $4x + 3y + 2z = 2009$ and $4x + 3y + 2z = 2000$, respectively. Find $(m - n) \bmod 1000$.

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2009 AIME II Problem 12: Distinct pair sums with disjoint endpoints

Choose disjoint pairs of distinct elements from $\{1, \ldots, 2009\}$, requiring all pair sums distinct and at most 2009. Find the greatest number of pairs.

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2010 AIME I Problem 1: One square divisor in a pair

Choose two distinct positive divisors of 2010^2 uniformly. Find numerator plus denominator of the probability that exactly one is a square.

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2010 AIME I Problem 4: Collision probability of head counts

Two fair coins and one coin with head probability $4/7$ are flipped independently by each of two people. Find numerator plus denominator of the probability that their head counts agree.

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2010 AIME I Problem 7: Exactly three pairwise intersection elements

Count ordered triples (A, B, C) of subsets of a seven-element set such that each pairwise intersection has size one and the triple intersection is empty. Report the residue modulo 1000.

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2010 AIME I Problem 10: Digits allowed to carry

Count representations $2010 = 1000a_3 + 100a_2 + 10a_1 + a_0$ with every integer a_i between 0 and 99.

Published problem and solutions · MAA AMC.

2010 AIME I Problem 12: A multiplicative two-color threshold

Find the least $m \geq 3$ such that every two-coloring of $\{3, \ldots, m\}$ has a monochromatic solution $ab = c$, allowing $a = b$.

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2010 AIME II Problem 2: Distance to the nearest side

Choose a point uniformly inside a unit square. Its distance to the nearest side is d. Find numerator plus denominator of $P(1/5 \leq d \leq 1/3)$.

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2010 AIME II Problem 3: Valuation of all pair differences

Let $K = \prod_{1 \leq a < b \leq 20}(b - a)$. Find the largest exponent n for which $2^n \mid K$.

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2010 AIME II Problem 4: A short gate change

Twelve gates are equally spaced 100 feet apart. A traveler is assigned one uniformly, then reassigned uniformly to a different gate. Find numerator plus denominator of the probability of walking at most 400 feet.

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2010 AIME II Problem 8: A set cannot contain its own size

Partition $\{1, \ldots, 12\}$ into ordered nonempty sets A, B, requiring $|A| \notin A$ and $|B| \notin B$. Count the partitions.

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2010 AIME II Problem 10: Quadratics modulo interchange of roots

Count quadratic polynomials with integer coefficients and integer roots whose constant coefficient is 2010.

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2010 AIME II Problem 11: Tic-tac-toe line counts

Count 3×3 binary boards with five ones and four zeros such that at most one row, column, or long diagonal is monochromatic.

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2010 AIME II Problem 13: Two fixed cards share a rank team

Four people draw distinct uniform cards from $1, \ldots, 52$ and the two lowest form one team. Two specified people hold $a, a + 9$. Among choices with their same-team probability at least $1/2$, find numerator plus denominator of the smallest probability.

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2011 AIME I Problem 5: Sliding sums force periodic residues

Label a regular nonagon by $1, \ldots, 9$, requiring every three consecutive labels to have sum divisible by 3. Identify rotations but not reflections. Count arrangements.

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2011 AIME I Problem 7: Splitting a power into equal smaller powers

For how many positive integers m can one power m^{x_0} equal a sum of exactly 2011 powers m^{x_i} with all exponents nonnegative?

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2011 AIME I Problem 10: Obtuse triangles in a regular polygon

Three vertices are chosen uniformly from a regular n-gon. Their triangle is obtuse with probability $93/125$. Find the sum of all possible n.

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2011 AIME I Problem 12: Conditioning on no isolated men

Six men and w women stand uniformly in a row. Given that no man is isolated, find the least w for which the probability of a male run of length at least four is at most $1/100$.

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2011 AIME II Problem 6: Reflect the outer gaps

Count $1 \le a < b < c < d \le 10$ with $a + d > b + c$.

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2011 AIME II Problem 7: Transitions bound the number of marbles

Arrange five identical green and some identical red marbles in a row. The number of equal-color adjacencies equals the number of unequal-color adjacencies. At the maximum possible red count, count arrangements modulo 1000.

Published problem and solutions · MAA AMC.

2011 AIME II Problem 12: Avoid a monochromatic triple on a circle

Seat three delegates from each of three countries randomly around a nine-seat circle. Every delegate must have a neighbor from another country. Find numerator plus denominator of the success probability.

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2011 AIME II Problem 14: Permutations respecting residue partitions

Count permutations $a_1, \dots, a_{30}$ of $1, \dots, 30$ satisfying $a_{i+m} \equiv a_i \pmod{m}$ for $m = 2, 3, 5$ wherever defined. Give the residue modulo 1000.

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2011 AIME II Problem 15: Floor equality on three intervals

Let $P(x) = x^2 - 3x - 9$ and choose x uniformly on $[5, 15]$. The probability that $\lfloor \sqrt{P(x)} \rfloor = \sqrt{P(\lfloor x \rfloor)}$ has form $(\sqrt{a} + \sqrt{b} + \sqrt{c} - d)/e$. Find $a + b + c + d + e$.

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2012 AIME I Problem 1: Divisibility in both digit directions

Count three-digit integers abc with $a, c \neq 0$ such that both abc and cba are divisible by four.

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2012 AIME I Problem 3: Exactly one correct meal type

Nine labeled diners order three meals of each of three types. The nine meal types are assigned to diners, with identical meals indistinguishable. Count assignments with exactly one diner receiving the ordered type.

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2012 AIME I Problem 5: Consecutive binary numbers with equal weight

Let B contain all thirteen-bit strings with eight ones and five zeros, allowing leading zeros. Count ordered pairs $(u, v) \in B^2$ with numeric difference $u - v = 1$.

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2012 AIME I Problem 7: Stationary coins on a network

On a connected undirected network, one central vertex has degree five, five vertices have degree three, and ten have degree four. The vertices hold 3360 coins and simultaneously split all coins equally among neighbors; every holding remains unchanged. Find the central holding.

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2012 AIME I Problem 11: Reachability in a lattice semigroup

From $(0, 0)$ allow jumps $(7, 2), (2, 7), (-5, -10), (-10, -5)$. Let M count reachable lattice points with $|x| + |y| \leq 100$. Find $M \bmod 1000$.

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2012 AIME I Problem 15: Multiplication changes every circular separation

Reseat n people on a numbered circle by $k \mapsto ak \pmod{n}$. Require this to be a permutation and require every pair's circular separation to change even up to reversal. Count possible $1 < n < 1000$.

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2012 AIME II Problem 3: Department and gender margins

Three departments each contain two men and two women. Choose a six-person committee with two members per department and three of each gender. Count committees.

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2012 AIME II Problem 7: Unranking a binary subset

List positive integers with exactly eight binary ones in increasing order. Find the 1000th integer modulo 1000.

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2012 AIME II Problem 12: Independent forbidden residues

A positive integer is p-safe if its distance from every multiple of p exceeds two. Count integers at most 10000 that are simultaneously 7-, 11-, and 13-safe.

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2012 AIME II Problem 14: Labeled two-regular graphs

Nine labeled people each shake hands with exactly two others. Count possible sets of handshakes modulo 1000.

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2013 AIME I Problem 2: Uniform digit sums modulo five

Count five-digit integers divisible by five whose first and last digits agree and whose digit sum is divisible by five.

Published problem and solutions · MAA AMC.

2013 AIME I Problem 4: Coloring the orbits of a quarter-turn

A thirteen-cell figure has one central cell and three four-cell orbits under a quarter-turn. Choose eight cells red and five blue uniformly. The probability of quarter-turn invariance is $1/n$. Find n.

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2013 AIME I Problem 6: Track only the special books

Randomly pack twelve distinct books into labeled boxes of sizes three, four, and five. Three books are mathematics books. Find numerator plus denominator of the probability that all three share a box.

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2013 AIME I Problem 15: Two arithmetic progressions with reordered residues

Count $0 \le A < B < C \le 99$ forming an arithmetic progression for which a prime p and residues $0 \le b < a < c < p$ satisfy $A \equiv a$, $B \equiv b$, $C \equiv c \pmod{p}$, with (b, a, c) also an arithmetic progression.

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2013 AIME II Problem 9: Colored compositions use all colors

Tile a seven-cell row by positive integer-length tiles colored red, blue, or green, requiring all three colors. Adjacent tiles of the same color remain distinct tiles. Find the tiling count modulo 1000.

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2013 AIME II Problem 11: Functions whose second iterate is constant

Count functions on a seven-element labeled set for which $f \circ f$ is constant, and give the count modulo 1000.

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2013 AIME II Problem 12: Cubics with roots on two circles

Count monic integer-coefficient cubics all of whose complex roots have modulus either 13 or 20.

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2014 AIME I Problem 2: Matching colors from two urns

Two independent draws use urns with color counts $(4, 6)$ and $(16, N)$. Their colors agree with probability $29/50$. Determine N.

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2014 AIME I Problem 5: Disk-separable vertex subsets

Count subsets of a regular dodecagon that are precisely its vertices strictly inside some circle.

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2014 AIME I Problem 11: Diagonal endpoint of a random walk

A planar simple random walk starts at the origin and takes six unit axis steps. Find numerator plus denominator of its probability of ending on either diagonal $y = x$ or $y = -x$.

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2014 AIME I Problem 12: Disjoint ranges of random functions

Choose independently two maps from a four-element set to itself. Find the numerator of the reduced probability that their images are disjoint.

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2014 AIME II Problem 2: Conditional probability from Venn atoms

Three events each have exclusive probability 0.10; every exclusive pair has probability 0.14. Given A and B, all three occur with probability $1/3$. Find numerator plus denominator of the probability of no events conditional on not A.

Published problem and solutions · MAA AMC.

2014 AIME II Problem 6: A hidden die and Bayesian updating

Choose equally between a fair die and one whose six-probability is $2/3$. After two sixes, find the reduced numerator plus denominator of the probability of another six.

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2014 AIME II Problem 9: A run of three on a cycle

Among the subsets of ten labeled cyclic seats, count those containing three consecutive seats.

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2014 AIME II Problem 13: Shoe matching and long cycles

Randomly match ten left shoes to ten right shoes. No subset of fewer than five resulting pairs may contain exactly the complete shoes of that many owners. Find numerator plus denominator of the probability.

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2015 AIME I Problem 2: Exactly one repeated nationality

Select three of nine distinct delegates, with nationality group sizes $2, 3, 4$. Find numerator plus denominator of the probability their nationality multiplicities are 2 and 1.

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2015 AIME I Problem 5: First matching sock pair on day three

Ten socks form five color pairs. Draw two without replacement on each of three days. Find numerator plus denominator of the probability only the third day produces a matching pair.

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2015 AIME I Problem 9: When a multiplicative recurrence reaches zero

For $1 \le x, y, z \le 10$ initialize $a_1 = x, a_2 = y, a_3 = z$ and $a_n = a_{n-1}|a_{n-2} - a_{n-3}|$. Count initial triples whose sequence eventually reaches zero.

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2015 AIME I Problem 12: Mean minimum of a large subset

Choose uniformly a 1000-element subset of $\{1, \dots, 2015\}$. Find numerator plus denominator of its expected minimum.

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2015 AIME I Problem 14: Parity of piecewise-linear area

Let $A(n)$ be the area under $y = x\lfloor\sqrt{x}\rfloor$ from $x = 1$ to $x = n$. Count $2 \le n \le 1000$ for which $A(n)$ is integral.

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2015 AIME II Problem 2: Conditioning on enrollment

Class fractions are $0.40, 0.30, 0.20, 0.10$; Latin participation rates respectively are $1, 0.80, 0.50, 0.20$. Given a Latin student, find numerator plus denominator of the probability of second-year status.

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2015 AIME II Problem 5: Adjacent cells in a random pair

Two different cells are chosen uniformly from an n by n square grid. Find the least n for which their probability of sharing an edge is below $1/2015$.

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2015 AIME II Problem 10: Permutations with bounded downward steps

Count permutations of 1 through 7 in which $a_k - a_{k+1} \leq 2$ for every adjacent pair.

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2015 AIME II Problem 12: Binary strings with bounded runs

Count length-ten binary words whose constant runs have length at most three.

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2016 AIME I Problem 2: Weighted dice

Each independent die has $P(k) = k/21$ for $k = 1, \ldots, 6$. Find numerator plus denominator of the probability their sum is seven.

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2016 AIME I Problem 3: Downward simple paths on an icosahedron

Orient an icosahedron with top and bottom vertices and two horizontal pentagons. Count vertex-simple edge paths from top to bottom that never move upward.

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2016 AIME I Problem 8: Minimizing a digit-column sum

Permute digits $1, \ldots, 9$ into three three-digit numbers. Among sums ending in 0, let m be the least sum and n its number of permutations. Find $|m - n|$.

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2016 AIME I Problem 13: Reflected random walk hitting time

A frog starts at height 21. Above 0 it chooses four unit axis moves uniformly; at height 0 only the two horizontal moves and upward move are allowed. Find expected steps to reach height 24.

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2016 AIME I Problem 14: Lattice shapes intersected by a rational line

At every lattice point place a radius 1/10 circle and an axis-parallel side 1/5 square. Count the total shapes meeting the segment from $(0, 0)$ to $(1001, 429)$.

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2016 AIME II Problem 2: Dependent rain events

Saturday rain has probability 2/5 and Sunday rain 3/10. Sunday rain is twice as likely after Saturday rain as after no Saturday rain. Find numerator plus denominator for rain on at least one day.

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2016 AIME II Problem 4: Layer counts determine a colored box

An $a \times b \times c$ box of unit cubes has 9 red and 12 green cubes in each layer perpendicular to the a-direction, and 20 green and 25 yellow cubes in each layer perpendicular to the b-direction. Find its minimum volume.

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2016 AIME II Problem 6: Absolute coefficient sum by parity

For $P(x) = 1 - x/3 + x^2/6$ and $Q(x) = P(x)P(x^3)P(x^5)P(x^7)P(x^9)$, find numerator plus denominator of the sum of absolute coefficients of Q.

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2016 AIME II Problem 8: Three distinct factors of a product

Count unordered triples of distinct positive integers whose product is $11 \cdot 21 \cdot 31 \cdot 41 \cdot 51 \cdot 61$.

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2016 AIME II Problem 11: Divisor counts of perfect powers

Call N k-nice when some kth power has exactly N positive divisors. Count $1 \le N < 1000$ that are neither 7-nice nor 8-nice.

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2016 AIME II Problem 12: Proper colorings of a six-cycle

Six fixed cyclic regions are colored with four colors; adjacent regions must differ. Count colorings.

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2016 AIME II Problem 13: Mean minimum score of a permutation board

Place six nonattacking rooks on a labeled 6 by 6 board. Each square scores row+column. Find numerator plus denominator of the mean minimum occupied score.

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2017 AIME I Problem 1: Triangles from boundary points

A triangle has 3 additional points on one side, 4 on another, and 5 on the third. Count nondegenerate triangles whose vertices are among all 15 marked points.

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2017 AIME I Problem 6: A chord intersecting an isosceles triangle

An inscribed isosceles triangle has equal angles x degrees. A chord with two independent uniform endpoints intersects the triangle with probability 14/25. Find the difference between possible x values.

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2017 AIME I Problem 7: A triple Vandermonde sum

Evaluate modulo 1000 the sum of $C(6,a)C(6,b)C(6,a+b)$ over $a,b \geq 0, a+b \leq 6$.

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2017 AIME I Problem 8: Two random right-triangle vertices

On the same semicircle of diameter $OP = 200$, select Q, R so that angles POQ and POR are independently uniform between 0 and 75 degrees. Find numerator plus denominator of $P(QR \leq 100)$.

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2017 AIME I Problem 11: Median of row medians

Arrange $1, \ldots, 9$ in a 3 by 3 array. Count modulo 1000 arrangements whose median of the three row medians is 5.

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2017 AIME I Problem 12: Product-free subsets

Count subsets S of $\{1, \ldots, 10\}$ containing no a, b, c (possibly equal) with $ab = c$.

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2017 AIME II Problem 1: Subsets avoiding two containers

Count subsets of [8] contained in neither $\{1,2,3,4,5\}$ nor $\{4,5,6,7,8\}$.

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2017 AIME II Problem 2: Weighted tournament champion

Four teams indexed $1, \ldots, 4$ play semifinals 1 versus 4 and 2 versus 3. Team i beats team j with probability $i/(i+j)$, independently by match. Find numerator plus denominator for team 4 winning.

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2017 AIME II Problem 3: Voronoi region inside a triangle

A point is uniform in the triangle with vertices $(0,0), (12,0), (8,10)$. Find numerator plus denominator of its probability of being closest to $(12,0)$.

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2017 AIME II Problem 4: Ternary digit avoidance

Count positive integers at most 2017 with no zero digit in base 3.

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2017 AIME II Problem 9: An eight-edge bipartite covering

Choose 8 cells of a 7 by 7 board uniformly, conditional on hitting every row and column. Find numerator plus denominator of the probability one cell can be removed while still hitting all rows and columns.

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2017 AIME II Problem 11: Strong tournaments on five vertices

Orient every edge of K_5. Count orientations in which every vertex can reach every other vertex.

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2017 AIME II Problem 13: Isosceles triangles in regular polygons

Let $f(n)$ count isosceles vertex triangles of a regular n-gon, including equilateral ones. Find the sum of $n \geq 3$ satisfying $f(n+1) - f(n) = 78$.

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2017 AIME II Problem 14: Lines containing exactly eight lattice points

In $\{1, \ldots, 10\}^3$ count lines containing exactly 8 grid points.

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2018 AIME I Problem 1: Integer-root quadratics counted by a sum

Count modulo 1000 pairs (a, b) with $1 \leq a \leq 100, b \geq 0$ for which $x^2 + ax + b$ splits over the integers.

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2018 AIME I Problem 3: Contiguous color blocks in a sample

Randomly order five cards drawn without replacement from five red and five green cards. Find numerator plus denominator of the chance that each color appears in at most one block.

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2018 AIME I Problem 7: Isosceles triangles in a hexagonal prism

A right regular hexagonal prism has base side 1 and height 2. Count isosceles triangles among its 12 vertices.

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2018 AIME I Problem 9: Two prescribed pair sums

Count four-element subsets of [20] having a pair summing 16 and a pair summing 24.

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2018 AIME I Problem 10: Directed walk on two pentagonal rings

Two five-vertex cycles have opposite one-way orientations and bidirectional matching spokes. Count modulo 1000 fifteen-step walks returning to a fixed start.

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2018 AIME I Problem 12: Subset sums modulo three

Choose a subset of [18] uniformly. Find the numerator of its reduced probability of having sum divisible by 3.

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2018 AIME I Problem 14: First arrival on a heptagon

A walk starts at S on a seven-cycle whose target E is three edges away in one direction and four in the other. Count walks reaching E for the first time within 12 steps.

Published problem and solutions · MAA AMC.

2018 AIME II Problem 6: Random parameter with four real roots

For a chosen uniformly in $[-20, 18]$, find numerator plus denominator of the probability that $x^4+2ax^3+(2a-2)x^2+(-4a+3)x-2$ has only real roots.

Published problem and solutions · MAA AMC.

2018 AIME II Problem 8: Axis jumps of sizes one and two

Count paths from $(0,0)$ to $(4,4)$ using steps $(1,0), (2,0), (0,1), (0,2)$.

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2018 AIME II Problem 10: Functions whose second image is fixed

Count maps $f : [5] \to [5]$ satisfying $f^2 = f^3$.

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2018 AIME II Problem 11: Indecomposable permutations

Count permutations of [6] such that no proper initial block contains precisely its smallest k labels.

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2018 AIME II Problem 13: Parity of waiting for a die pattern

Roll a fair die until 123 first appears in consecutive rolls. Find numerator plus denominator of the chance the stopping time is odd.

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2018 AIME II Problem 15: Integer bi-Lipschitz sequences

Count integer-valued functions on $\{0, \ldots, 6\}$ with $f(0) = 0, f(6) = 12$ and $|x-y| \leq |f(x)-f(y)| \leq 3|x-y|$.

Published problem and solutions · MAA AMC.

2019 AIME I Problem 2: A gap between ordered samples

Choose distinct J, B uniformly from [20] in order. Find numerator plus denominator of $P(B - J \geq 2)$.

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2019 AIME I Problem 4: Ordered substitutions

A team has 11 active players and 11 unused substitutes. Count modulo 1000 histories with at most 3 successive substitutions, with removed players unable to return.

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2019 AIME I Problem 5: Hitting the corner before either axis

From $(4, 4)$, independently subtract $(1, 0), (0, 1)$, or $(1, 1)$, each with probability $1/3$, until an axis is reached. Write the chance of first hitting $(0, 0)$ as $m/3^n$ with 3 not dividing m. Find $m + n$.

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2019 AIME II Problem 2: Hitting a site with jumps one and two

A frog starts on 1 and independently advances 1 or 2 with equal probabilities. Find numerator plus denominator of its chance of visiting 7.

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2019 AIME II Problem 3: Overlapping products of seven integers

Count positive integer tuples (a, b, c, d, e, f, g) satisfying $abc = 70, cde = 71, efg = 72$.

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2019 AIME II Problem 4: A square product of four die rolls

Find numerator plus denominator of the probability that four fair die outcomes have square product.

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2019 AIME II Problem 5: Ambassadors with adjacent advisors

Four distinct ambassador-advisor pairs occupy 12 labeled cyclic chairs. Ambassadors use even chairs; each advisor must neighbor their ambassador. Count modulo 1000 valid seatings.

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2019 AIME II Problem 12: Divisibility chains with prescribed sum

Count finite strictly increasing positive integer sequences, each term dividing the next, whose sum is 360.

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2019 AIME II Problem 14: A three-denomination Frobenius problem

With unlimited stamps of denominations $5, n, n+1$, the largest unattainable value is 91. Find the sum of possible n.

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2020 AIME I Problem 5: One deletion makes a permutation monotone

Count permutations of [6] from which deleting one entry leaves an increasing or decreasing sequence.

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2020 AIME I Problem 7: A balanced committee bijection

Choose a committee from 11 men and 12 women with exactly one more woman than man. Find the sum of distinct prime divisors of the number of committees.

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2020 AIME I Problem 9: A random chain of divisors

Independently sample three divisors a, b, c of 20^9 uniformly with replacement. Find the reduced numerator of $P(a \mid b \text{ and } b \mid c)$.

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2020 AIME I Problem 11: Counting coefficients through common images

For integer a, b, c in $[-10, 10]$, set $f(x) = x^2 + ax + b$ and $g(x) = x^2 + cx + d$. Count triples for which an integer d makes $g(f(2)) = g(f(4)) = 0$.

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2020 AIME II Problem 2: A random slope in a square

Choose P uniformly in the unit square. Find numerator plus denominator of the chance that its slope from $(5/8, 3/8)$ is at least $1/2$.

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2020 AIME II Problem 9: Avoiding former neighbors

Six people formerly sat in order 123456. Count new orders in which no formerly adjacent pair is adjacent.

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2020 AIME II Problem 12: A line crossing a numbered grid

An odd m by odd n row-major grid has label 200 in its top row and 2000 in its bottom row. Count (m, n) for which the line through those cell centers enters cell 1099.

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2020 AIME II Problem 14: Iterated fractional-part preimages

Define $f(x) = x\{x\}$, where $\{x\} = x - \lfloor x \rfloor$. Count modulo 1000 the solutions of $f(f(f(x))) = 17$ in $[0, 2020]$.

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2021 AIME I Problem 1: One loss in a persistent two-state process

Zou wins race 1. Thereafter the previous winner wins the next race with probability $2/3$. Find numerator plus denominator of the chance Zou wins exactly 5 of 6 races.

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2021 AIME I Problem 3: Differences of powers of two

Count positive integers below 1000 expressible as a difference of integral powers of 2.

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2021 AIME I Problem 4: Three strictly increasing piles

Count integer triples $0 < a < b < c$ with $a + b + c = 66$.

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2021 AIME I Problem 7: Simultaneous sine maxima

Count $1 \le m < n \le 30$ for which some real x satisfies $\sin(mx) + \sin(nx) = 2$.

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2021 AIME I Problem 8: Counting levels with twelve preimages

Count integers c for which $||20|x| - x^2| - c| = 21$ has 12 distinct real solutions.

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2021 AIME I Problem 12: Three frogs until their first collision

Three independent symmetric walkers begin equally spaced on a 12-cycle. They stop when two occupy the same vertex. Find numerator plus denominator of their expected stopping time.

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2021 AIME II Problem 1: Mean of three-digit palindromes

Find the arithmetic mean of all decimal three-digit palindromes.

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2021 AIME II Problem 3: Cyclic products modulo three

Permute $1, \ldots, 5$ around five labeled positions. Count permutations for which the sum of the five products of three cyclically consecutive entries is divisible by 3.

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2021 AIME II Problem 6: An equality characterizing nested sets

Count ordered subset pairs (A, B) of $[5]$ with $|A||B| = |A \cap B||A \cup B|$.

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2021 AIME II Problem 8: Nonbacktracking walk on a cube

On a cube, an ant starts at a bottom vertex, chooses its first edge uniformly, and thereafter uniformly chooses either edge that does not reverse its previous step. Find numerator plus denominator of its chance of being on top after 8 steps.

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2021 AIME II Problem 9: Counting exponent pairs by valuations

Count ordered $1 \leq m, n \leq 30$ with $\gcd(2^m + 1, 2^n - 1) > 1$.

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2021 AIME II Problem 11: Learning from simultaneous ignorance

Four logical students each know one member of an unknown set of four consecutive two-digit integers. The set has different multiples of 6 and 7. Initially none can determine it; hearing this, all can. Find the sum of possible maximum elements.

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2022 AIME I Problem 4: Roots-of-unity exponent pairs

Let $w = (\sqrt{3} + i)/2$ and $z = (-1 + i\sqrt{3})/2$. Count $1 \leq r, s \leq 100$ with $iw^r = z^s$.

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2022 AIME I Problem 6: Avoiding four-term progressions

Count choices $5 < a < b < 30$ such that $\{3, 4, 5, a, b, 30, 40, 50\}$ contains no four-term arithmetic progression.

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2022 AIME I Problem 7: A minimal positive ratio of digit products

Partition digits 1 through 9 into three ordered triples. Find numerator plus denominator of the least positive value $(abc - def)/(ghi)$.

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2022 AIME I Problem 9: Two copies in opposite parity positions

Randomly arrange two indistinguishable blocks of each of six colors. Find numerator plus denominator of the probability that every equal-color pair has an even number of intervening blocks.

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2022 AIME I Problem 12: Summing intersection sizes

Let S_n sum $|A \cap B|$ over ordered equal-sized subset pairs of $[n]$. If $S_{2022}/S_{2021} = p/q$ in lowest terms, find $(p+q) \pmod{1000}$.

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2022 AIME I Problem 13: Distinct reduced numerators

Consider all nonzero repeating four-digit decimals. Count their distinct reduced numerators modulo 1000.

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2022 AIME II Problem 2: A randomly seeded tournament

Four tennis players include A, C, J, S. C beats A with probability $1/3$; A and C each beat J or S with probability $3/4$. Semifinal pairings are uniform. Find numerator plus denominator for C becoming champion.

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2022 AIME II Problem 5: Prime-difference triangles

Twenty cyclic vertices are labeled $1, \ldots, 20$; connect labels whose ordinary difference is prime. Count triangles with original vertices.

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2022 AIME II Problem 8: Identifiability from three floor values

Count positive integers $n \leq 600$ uniquely determined among positive integers by $\lfloor n/4 \rfloor, \lfloor n/5 \rfloor, \lfloor n/6 \rfloor$.

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2022 AIME II Problem 9: Regions between two supporting lines

Join all m marked points on one line to all n on a parallel line, with no triple interior intersection. Including the supporting lines, count bounded regions for $m = 7, n = 5$.

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2022 AIME II Problem 10: Choosing two edges

Evaluate modulo 1000 the sum from $k = 3$ to 40 of $\binom{\binom{k}{2}}{2}$.

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2022 AIME II Problem 13: A coefficient reduced to stars and bars

Find coefficient x^{2022} in $(x^{2310}-1)^6/[(x^{105}-1)(x^{70}-1)(x^{42}-1)(x^{30}-1)]$.

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2022 AIME II Problem 14: Complete coverage by a finite stamp collection

For denominations $a < b < c$, let f be the minimum number of stamps, using every denomination, whose subcollections represent every value from 1 through 1000. Find and sum the three least values of c permitting $f = 97$.

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2023 AIME I Problem 1: No opposite pair of men

Five men and nine women stand randomly at 14 cyclic positions. Find numerator plus denominator of the probability every man is opposite a woman.

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2023 AIME I Problem 3: Counting ordinary intersections

Forty lines have no parallel pair. There are exactly 3 triple points, 4 quadruple points, 5 quintuple points, and 6 sextuple points, and no higher multiplicity. Count double points.

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2023 AIME I Problem 6: Optimal sequential color guesses

Three red and three black cards are revealed in random order. Before each reveal, guess the color using previous observations. Find numerator plus denominator of the optimal expected correct guesses.

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2023 AIME I Problem 7: Pairwise-distinct remainders

Count positive $n < 1000$ whose remainders modulo $2, 3, 4, 5, 6$ are all different.

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2023 AIME I Problem 9: A cubic with one other equal-value integer

Count monic cubics $x^3 + ax^2 + bx + c$ with a, b, c in $[-20, 20]$ for which exactly one integer m unequal 2 satisfies $p(m) = p(2)$.

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2023 AIME I Problem 11: Exactly one adjacent selected pair

Count subsets of [10] containing exactly one pair of consecutive integers.

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2023 AIME I Problem 14: Hamiltonian cycles of two clock hands

Two independently advancing clock hands start at 12. Each move advances exactly one hand one step. Count modulo 1000 the 144-move tours visiting every ordered hand-position pair once before returning.

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2023 AIME II Problem 6: Midpoints in an L-shaped region

Choose two independent uniform points in a 2 by 2 square with its upper-right unit square removed. Find numerator plus denominator of the chance their midpoint remains in the region.

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2023 AIME II Problem 7: Avoiding monochromatic rectangles

Color 12 cyclic vertices red or blue. Count colorings with no monochromatic rectangle among the vertices.

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2023 AIME II Problem 10: Balanced proper colorings of a ladder

Place $1, \ldots, 12$ in a 2 by 6 grid so adjacent cells never have equal residues modulo 3. Let N count placements. Find the number of positive divisors of N.

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2023 AIME II Problem 11: Maximum intersecting families on five elements

Count collections of 16 distinct pairwise-intersecting subsets of [5].

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2024 AIME I Problem 3: Winning residues in a subtraction game

Two players alternately remove either 1 or 4 counters. The player taking the last counter wins. Count initial sizes from 1 through 2024 for which the second player can force a win.

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2024 AIME I Problem 4: The jackpot conditional on winning

A ticket chooses four of ten numbers; four winning numbers are drawn uniformly. A prize requires at least two matches. Find numerator plus denominator of the conditional probability of four matches, given a prize.

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2024 AIME I Problem 6: Grid paths with four turns

Count paths from one corner of an 8×8 grid to the opposite corner, using only right and upward steps, with exactly four changes of direction.

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2024 AIME I Problem 11: Rotating blue vertices away from themselves

Color the eight vertices of a regular octagon independently red or blue. Find numerator plus denominator of the probability that some rotation sends every blue vertex to a red vertex.

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2024 AIME II Problem 1: Recovering a fourfold intersection

Among 900 residents everyone has a candy. Ring, club, and spade ownership totals are $195, 367$, and 562. Exactly 437 residents own two of the four items and 234 own three. How many own all four?

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2024 AIME II Problem 3: Digits constrained by row and column sums

Fill a 2×3 array with decimal digits, allowing leading zeroes. The two three-digit rows sum to 999, and the three two-digit columns sum to 99. Count fillings.

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2024 AIME II Problem 6: Encoding a set by maxima of subsets

A finite set A of positive integers has the property that exactly 2024 finite nonempty sets of positive integers have maximum in A. Find the sum of the elements of A.

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2024 AIME II Problem 9: Maximal monochromatic rows and columns

Place black and white chips on a 5×5 board, at most one per cell, so every occupied row and column is monochromatic. Count placements to which no chip can be added.

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2024 AIME II Problem 11: A symmetric cubic counting problem

Count ordered nonnegative integer triples with $a+b+c = 300$ and $a^2b + a^2c + b^2a + b^2c + c^2a + c^2b = 6000000$.

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2024 AIME II Problem 15: Rectangles supported by dodecagon chords

Count rectangles whose four sides lie on sides or diagonals of a regular dodecagon. Rectangle corners need not be polygon vertices.

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2025 AIME I Problem 3: Flavor assignments with ordered popularity

Nine distinct people each choose one of three ice-cream flavors. Every flavor is chosen, with chocolate chosen more often than vanilla and vanilla more often than strawberry. Find the assignment count modulo 1000.

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2025 AIME I Problem 5: Digit permutations divisible by twenty-two

Let N count permutations of digits 1 through 8 that form an eight-digit integer divisible by 22. Find $N - 2025$.

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2025 AIME I Problem 7: The final alphabetized pair

Pair the twelve letters A through L uniformly into six unordered pairs. Sort each pair alphabetically and then sort the six pairs. Find numerator plus denominator of the probability the last pair contains G.

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2025 AIME I Problem 10: Counting Sudoku bands

Fill a 3×9 band with digits 1 through 9 so that every row and every consecutive 3×3 block contains each digit once. If the count is $2^a 3^b 5^c 7^d$, find $2a + 3b + 5c + 7d$.

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2025 AIME I Problem 13: Expected regions from random chords

Two perpendicular diameters divide a disk into four quadrants. Add 25 independent chords, each choosing uniformly a pair of distinct quadrants and then uniform endpoints within those quadrants. Find the expected number of regions.

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2025 AIME I Problem 15: Cubic residues and three-variable counting

Count ordered triples $1 \le a, b, c \le 729$ with $a^3 + b^3 + c^3$ divisible by 2187, and give the remainder modulo 1000.

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2025 AIME II Problem 3: Two red sides on every small square

Color the twelve unit edges of a 2×2 array of squares red or blue. Count colorings in which every small square has exactly two red edges.

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2025 AIME II Problem 7: The least common multiple of a random divisor subset

Choose uniformly a subset of the positive divisors of 2025. Find numerator plus denominator of the probability that its least common multiple is 2025.

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2025 AIME II Problem 8: When greedy coin change succeeds

With coin values $1, 10$, and 25, count amounts from 1 through 1000 for which repeatedly taking the largest affordable coin uses the fewest coins.

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2025 AIME II Problem 10: Eight occupied seats with no triple

Select eight of sixteen seats in a row so that no three consecutive seats are selected. Find the number of selections modulo 1000.

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2025 AIME II Problem 11: Equal-length perfect matchings on a polygon

Pair all vertices of a regular 24-gon so that the twelve resulting chords have the same length. Count the pairings.

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2026 AIME I Problem 2: Palindromes as compositions

Count positive decimal palindromes with no zero digits and digit sum 13.

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2026 AIME I Problem 7: Permutations whose sixth power is identity

Count permutations of six labeled elements whose sixth iterate fixes every element.

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2026 AIME I Problem 8: Divisor residues determined by exponent parity

Count divisors of 17017^{17} congruent to 5 modulo 12, and give the count modulo 1000.

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2026 AIME I Problem 9: Visible even stickers on a die

Roll a fair die six times, placing sticker k on the top face after roll k; a new sticker covers an old one. Given that stickers $2, 4$, and 6 remain visible, find numerator plus denominator of the chance exactly one face is blank.

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2026 AIME I Problem 11: Maximizing total variation on a grid

Place 1 through 64 bijectively in an 8×8 grid. Maximize the sum of absolute differences across all horizontal and vertical adjacencies; report the maximum modulo 1000.

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2026 AIME I Problem 13: Binomial residue sums over a finite field

For $0 \le r < 502$, let S_r sum $\binom{10000}{k}$ over $0 \le k \le 10000$ congruent to r modulo 502. How many S_r are divisible by 503?

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2026 AIME I Problem 15: Five rectangular cell loops

A cell loop consists of the boundary cells of an axis-aligned rectangle with both dimensions at least two. Count partitions of a 10×10 board into five cell loops.

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2026 AIME II Problem 2: Paths through a strip of diagonal squares

Ten adjacent squares each contain their rising diagonal. A path runs from the lower-left endpoint to the upper-right endpoint, never moving left and never reusing an edge. If there are N paths, find $\sqrt{N}$.

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2026 AIME II Problem 4: A numeral equal to its value in a smaller base

For positive decimal n, reinterpret its digit string in the base one larger than its greatest digit. Count $n < 1000$ unchanged by this operation.

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2026 AIME II Problem 5: Equal hypergeometric probabilities

An urn has at least seven marbles of each of two colors. Drawing seven without replacement gives equal probabilities of exactly four red marbles and exactly five red marbles. Sum the five smallest possible total populations.

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2026 AIME II Problem 7: Two successes of each kind before a stop

Independent trials choose A, B, or C with equal probabilities. If the chance that A and B each occur at least twice before the first C is m/n in lowest terms, find $100m + n$.

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2026 AIME II Problem 13: Sets with a prescribed number of adjacent partners

Finite integer sets S, T are cousins when they are disjoint, equally large, and admit a pairing with paired values differing by one. Find the smallest size of S having exactly 4040 cousins.

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2026 AIME II Problem 14: Compositions with parity-dependent subtraction

For integers a, b, define $a \circ b = a - b$ when a is odd and b even, and $a + b$ otherwise. Count positive-integer sequences of sum 12 whose left-associated operation value is zero.

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2026 AIME II Problem 15: A Fano-plane residue count

Count ordered seven-tuples with entries in $\{1, 2, 3\}$ for which both $\sum_i a_i$ and $\sum_i a_i a_{i+1} a_{i+3}$ are divisible by three; indices are cyclic modulo seven.

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